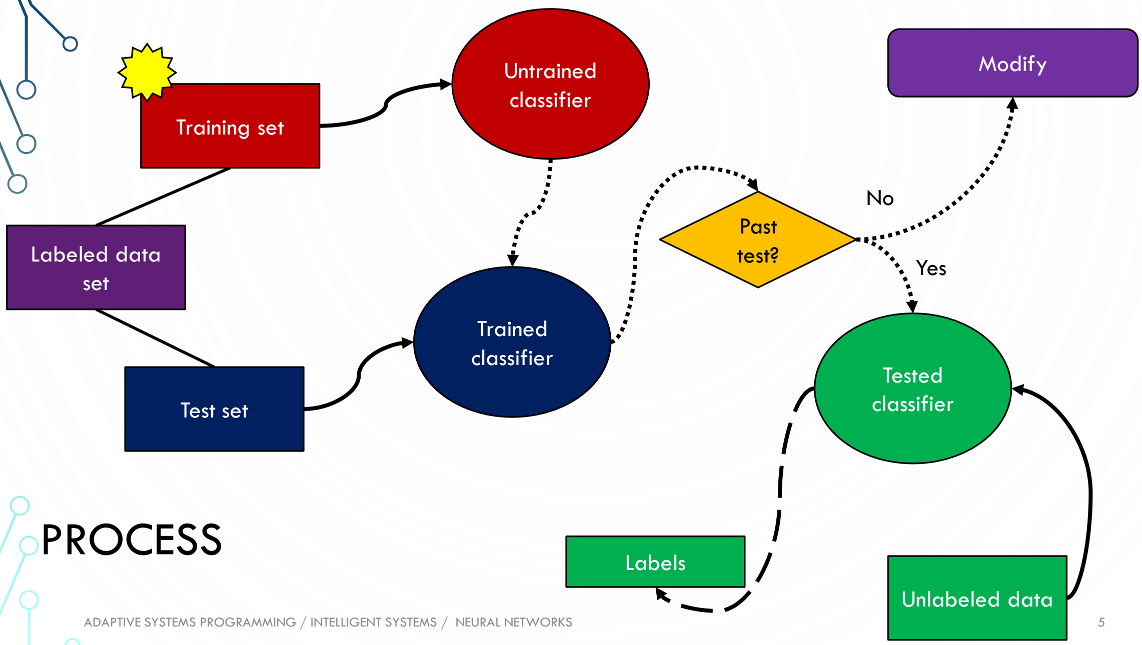
NEURAL NETWORKS

UNIT 3: INTELLIGENT SYSTEMS

SUPERVISED LEARNING



PROCESS

1 Training

Adjust parameter values of the classifier to accurately reproduce the labels of the training set.

2 Test

Check if classifier was trained correctly, i.e. that it can generalize and there was no overfitting.

3 Use

Utilize the classifier (generated model) to obtain labels for new data (not previously observed).

TRAINING SET

- Examples, objects, instances, feature vectors, inputs
- The labels of the examples are known.
- Used to find parameter values of the classifier.

TEST SET

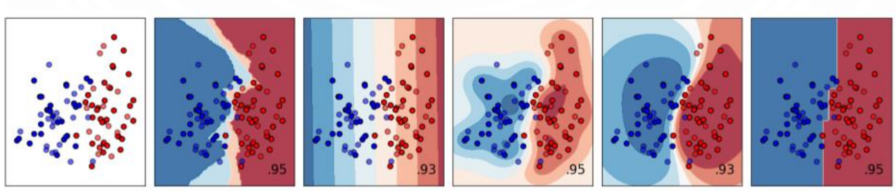
- Are also examples.
- Labels are also known for this set.
- Used for evaluation.

MODEL

- Set of **values** that correspond to the classifier's **parameters**.

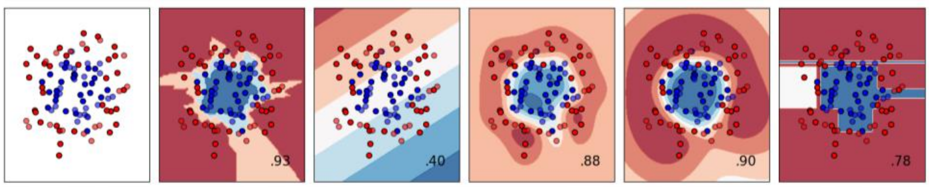
DECISION BOUNDARIES

- Classifiers, in general, aim to determine the boundaries among the different classes present in the data set.



DECISION BOUNDARIES

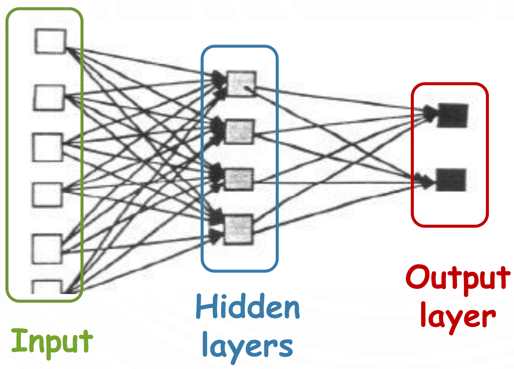
- Sometimes, boundaries are hard to determine.



GENERAL ARCHITECTURE

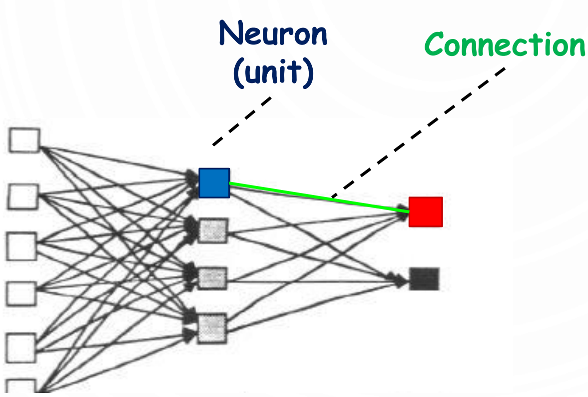
INPUTS / HIDDEN LAYERS / OUTPUT LAYER

LAYERS



- Receive **external inputs** (*examples*), processed with a series of layers.
- **Outputs** are produced in the last layer.
- Internal layers (that are not the output layer) are called ***hidden layers***.
- Inputs *are not considered* a layer.

LAYERS



- In each layer, we have a number of **neurons**.
- **Neuron** = processing unit
- Neurons in layer *i* are all connected to (all) neurons in layer *i+1*.
- Each connection has a **weight**.
- There are also special weights called **biases**.

- Training a neural network hence consists of **finding the weights** that allow to correctly classify the data set.
  - That is, that permit to recognize a pattern.

MULTIPLE NEURONS PER LAYER

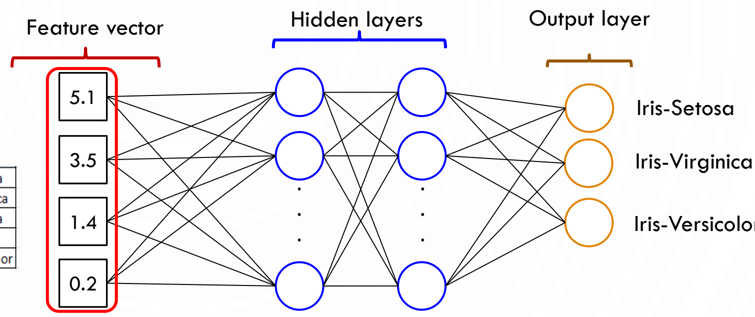
- Allow "parallel" processing

MULTIPLE LAYERS

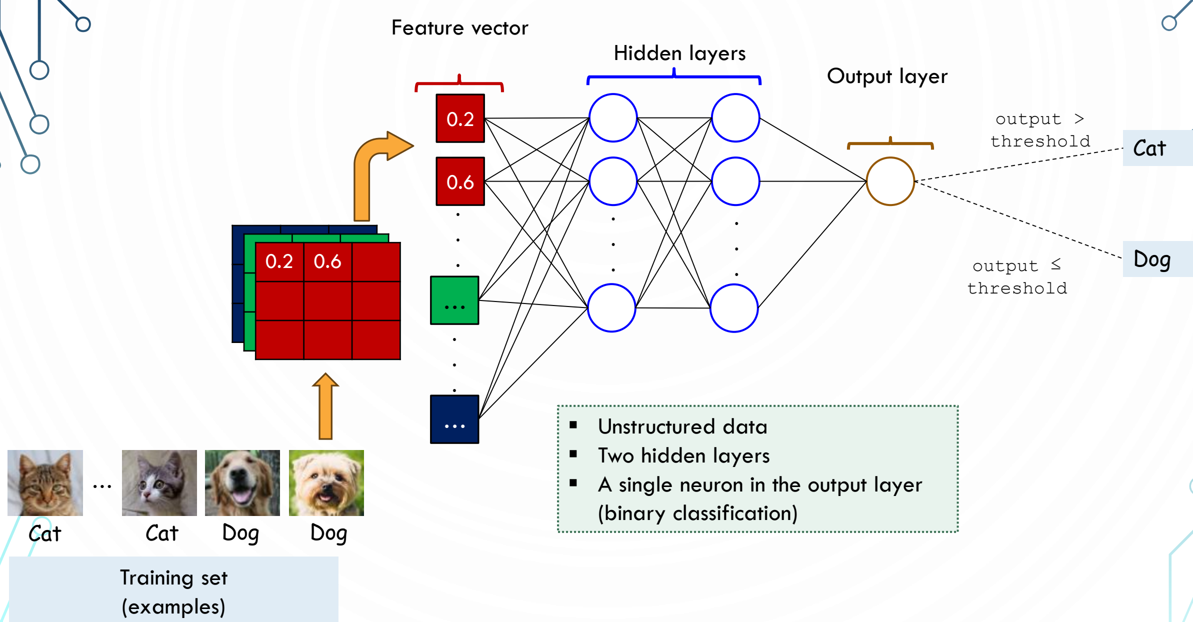
- Generate more sophisticated composite functions
- Shallow networks = 2 hidden layers, max.
- Deep networks = more than 2 hidden layers

Petal width	Sepal width	Petal length	Sepal length	Class
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3	1.4	0.2	Iris-virginica
4.7	3.2	1.3	0.2	Iris-setosa
...	...	...	...	...
5	3.6	1.4	0.2	Iris-versicolor

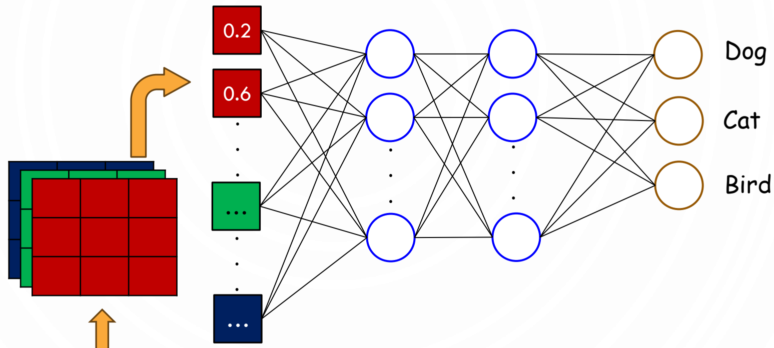
Training set
(examples)

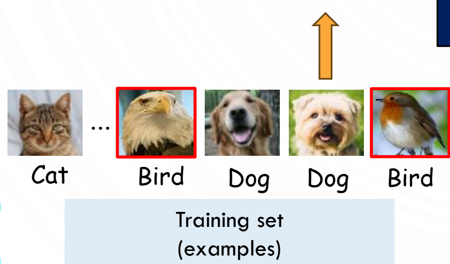


- Structured data
- Two hidden layers
- Several neurons in the output layer (multiclass classification)



- Unstructured data
- Two hidden layers
- A single neuron in the output layer (binary classification)

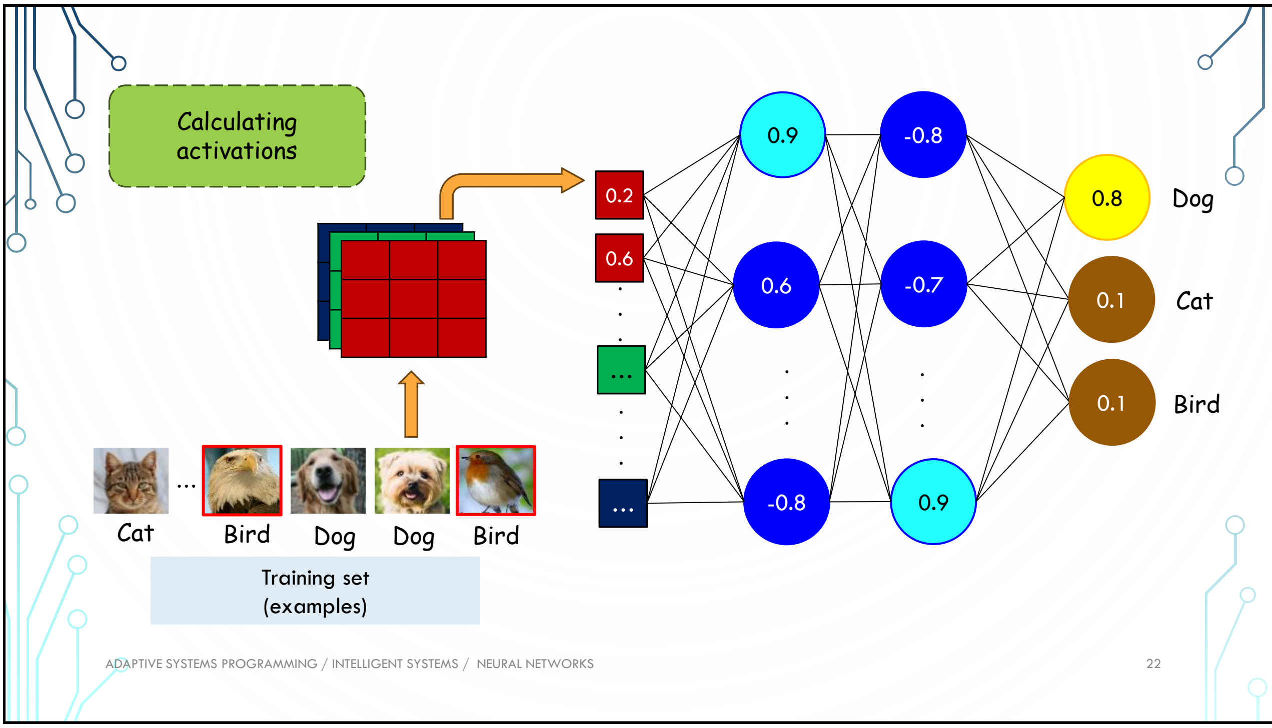


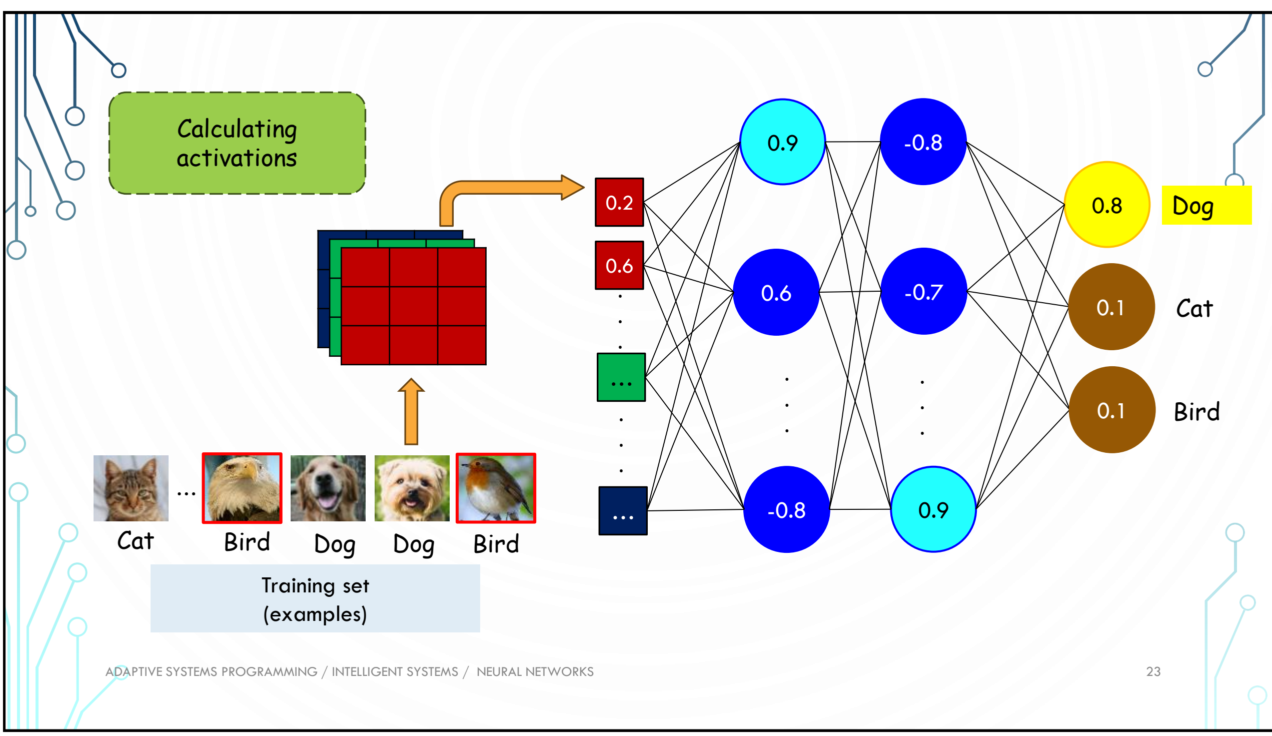


- Unstructured data
- Two hidden layers
- One neuron per class in the output layer (multiclass classification)

HOW DO THEY WORK?

ACTIVATIONS / WEIGHT UPDATE

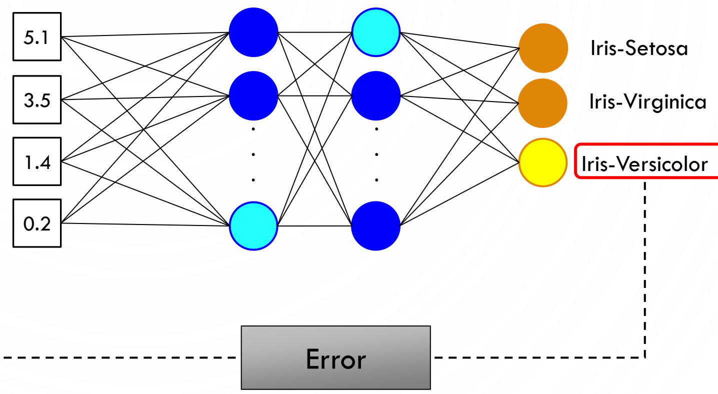

Calculating activations
0.2
0.6
...
...
0.9
0.6
-0.8
-0.8
-0.7
0.9
0.8
Dog
0.1
Cat
0.1
Bird
Cat
...
Bird
Dog
Dog
Bird
Training set (examples)
ADAPTIVE SYSTEMS PROGRAMMING / INTELLIGENT SYSTEMS / NEURAL NETWORKS
22


Calculating activations
0.2
0.6
...
...
0.9
0.6
-0.8
-0.8
-0.7
0.9
0.8
Dog
0.1
Cat
0.1
Bird
Cat
...
Bird
Dog
Dog
Bird
Training set (examples)
ADAPTIVE SYSTEMS PROGRAMMING / INTELLIGENT SYSTEMS / NEURAL NETWORKS
23

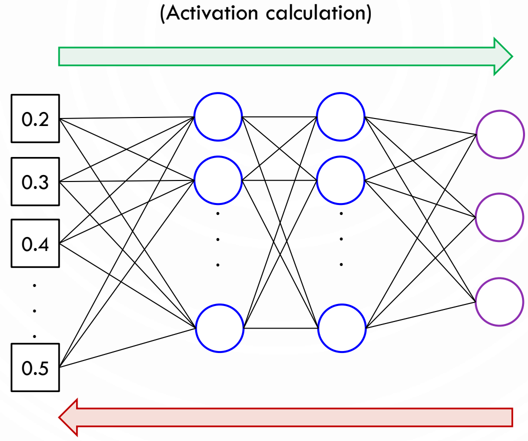
Updating weights

Petal width	Sepal width	Petal length	Sepal length	Class
5.1	3.5	1.4	0.2	Iris-setosa
4.9	3	1.4	0.2	Iris-virginica
4.7	3.2	1.3	0.2	Iris-setosa
...	...	...	...	
5	3.6	1.4	0.2	Iris-versicolor

Training set
(examples)



Forward propagation
(Activation calculation)



Back propagation
(Weight update)